

Introducing Google AI Edge Portal (<https://ai.google.dev/edge/ai-edge-portal>): Benchmark Edge AI at scale. Sign-up (<https://docs.google.com/forms/d/e/1FAIpQLSfTcGPycQve8TLAsfH46pBIXBZe9FrgJAClwbF7DeL1LgVn4Q/viewform>) to request access during private preview.

Run LLMs using LiteRT-LM

LiteRT-LM is a cross-platform library designed to efficiently run language model pipelines on a wide range of devices, from mobile phones to embedded systems. It provides developers with the tools to create and deploy sophisticated language model workflows, now with seamless NPU integration.

Run LLMs on CPU and GPU

See [LiteRT-LM GitHub repo](https://github.com/google-ai-edge/LiteRT-LM) (<https://github.com/google-ai-edge/LiteRT-LM>) for detailed instructions on cross-platform development and CPU/GPU hardware acceleration.

Run LLMs on NPU

Neural Processing Units (NPUs) offer specialized hardware blocks optimized for deep learning workloads. They are increasingly available in modern systems on a chip (SoCs), especially on mobile devices. Their high-performing nature makes them a great fit for running LLM inference.

NPU Vendors

LiteRT-LM supports running LLMs using NPU acceleration with the following vendors. Choose the instructions depending on which vendor you would like to try:

- [Google Tensor](#) (#tensor)
- [Qualcomm AI Engine Direct](#) (#qualcomm)
- [MediaTek NeuroPilot](#) (#mediatek)
- [Intel OpenVino](#) (#intel)

Quick Start

To get started, first follow the [Prerequisites](https://github.com/google-ai-edge/LiteRT-LM?tab=readme-ov-file#prerequisites)

(<https://github.com/google-ai-edge/LiteRT-LM?tab=readme-ov-file#prerequisites>) instructions to set up the environment and the repository.

Also, to be able to interact with your Android device, make sure you've properly installed [Android Debug Bridge](https://developer.android.com/tools/adb) (<https://developer.android.com/tools/adb>) and have a connected device that can be accessed using `adb`.

For more details instructions, checkout the [Quick Start](https://github.com/google-ai-edge/LiteRT-LM?tab=readme-ov-file#quick-start)

(<https://github.com/google-ai-edge/LiteRT-LM?tab=readme-ov-file#quick-start>) section in the [LiteRT-LM](https://github.com/google-ai-edge/LiteRT-LM)

(<https://github.com/google-ai-edge/LiteRT-LM>) repository and find more information about the [litert_lm_main command line demo](https://github.com/google-ai-edge/LiteRT-LM?tab=readme-ov-file#command-line-demo-usage-)

(<https://github.com/google-ai-edge/LiteRT-LM?tab=readme-ov-file#command-line-demo-usage->).

Google Tensor

Follow these steps to run LLMs on Google Tensor:

- ▶ **Step 1:** Download the `.litertlm` model
- ▶ **Step 2:** Build the LiteRT-LM runtime / libraries
- ▶ **Step 3:** Run the model on device

Qualcomm AI Engine Direct

The steps to run the LLMs on Qualcomm NPU are as the following:

- ▶ **Step 1:** Download the `.litertlm` model
- ▶ **Step 2:** Download and extract the QAI RT libraries
- ▶ **Step 3:** Build the LiteRT-LM runtime / libraries
- ▶ **Step 4:** Run the model on device

MediaTek NeuroPilot

The steps to run the LLMs on MediaTek NPU are as the following:

- ▶ **Step 1:** Download the `.litertlm` model
- ▶ **Step 2:** Build the LiteRT-LM runtime / libraries
- ▶ **Step 3:** Run the model on device

Intel OpenVino

Run using the C++ executable

The steps to run the LLMs on Intel NPU are as the following:

- ▶ **Step 1:** Download the `.litertlm` model
- ▶ **Step 2:** Build the LiteRT-LM runtime and LiteRT Intel Dispatch library
- ▶ **Step 3:** Run the model on device

Except as otherwise noted, the content of this page is licensed under the [Creative Commons Attribution 4.0 License](https://creativecommons.org/licenses/by/4.0/) (<https://creativecommons.org/licenses/by/4.0/>), and code samples are licensed under the [Apache 2.0 License](https://www.apache.org/licenses/LICENSE-2.0) (<https://www.apache.org/licenses/LICENSE-2.0>). For details, see the [Google Developers Site Policies](https://developers.google.com/site-policies) (<https://developers.google.com/site-policies>). Java is a registered trademark of Oracle and/or its affiliates.

Last updated 2026-06-12 UTC.